Machine Learning Classification of Heart and Lung Sound Recordings

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***Abstract*—This report presents a machine learning approach for classifying heart, lung, and mixed audio recordings. The project uses the HLS-CMDS dataset, which contains separate heart sound recordings, lung sound recordings, and mixed heart/lung recordings. The workflow includes audio preprocessing, feature extraction, model training, hyperparameter tuning, and performance evaluation. The final model is evaluated using classification metrics such as accuracy, precision, recall, F1-score, and confusion matrices. Replace this paragraph with a concise summary of the completed method, best result, and main finding.**

***Index Terms*—machine learning, audio classification, heart sounds, lung sounds, feature extraction, neural networks**

I. Introduction

The goal of this project is to classify biomedical audio recordings into meaningful sound categories using machine learning. Heart and lung sounds are clinically important signals, and automated classification can support faster screening, data organization, and diagnostic decision support.

Briefly introduce the problem, why it matters, and what this project attempts to solve. End the section with a short summary of the main contributions, such as:

- preprocessing and organizing the HLS-CMDS audio dataset;
- extracting useful audio features for classification;
- training and comparing machine learning models;
- evaluating model performance using standard classification metrics.

II. Dataset

The dataset used in this project is the HLS-CMDS dataset. It contains heart sound recordings, lung sound recordings, and mixed heart/lung recordings. The project directory is organized into the HS, LS, and Mix subsets.

TABLE I
Dataset Summary

Subset	Description	Recordings
HS	Heart sound recordings	50
LS	Lung sound recordings	50
Mix	Mixed heart/lung recordings	435

Describe the source of the data, the class labels, the number of samples per class, and any relevant characteristics of the audio files, such as sampling rate, duration, or class imbalance.

III. Preprocessing and Feature Extraction

Before model training, the audio recordings were preprocessed to produce a consistent input representation. This section should describe the steps used to clean, normalize, transform, or segment the audio data.

Possible preprocessing steps include:

- loading waveform files and labels;
- resampling audio to a consistent sampling rate;
- trimming or padding recordings to a fixed length;
- normalizing signal amplitude;
- converting audio into features such as MFCCs, spectrograms, or Mel-spectrograms.

If feature extraction was used, include the feature dimensions and explain why those features are appropriate for this classification task.

IV. Methodology

This section describes the machine learning models and training procedure used for the project. Include enough detail for another person to reproduce the experiment.

A. Model Architecture

Describe the model or models tested. For example, if a convolutional neural network was used, describe the convolutional layers, activation functions, pooling layers, fully connected layers, dropout, and output layer.

B. Training Procedure

Describe the train-validation-test split, loss function, optimizer, learning rate, batch size, number of epochs, and any hyperparameter tuning strategy.

C. Evaluation Metrics

The trained models were evaluated using standard classification metrics:

- accuracy;
- precision;
- recall;
- F1-score;
- confusion matrix.

TABLE II
MODEL PERFORMANCE COMPARISON

Model	Accuracy	Precision	Recall	F1-score
Baseline model	TODO	TODO	TODO	TODO
Tuned model	TODO	TODO	TODO	TODO
Final model	TODO	TODO	TODO	TODO

V. RESULTS

Present the main experimental results in this section. Include a table comparing the tested models or configurations.

Add the most important figures here, such as a confusion matrix, training loss curve, validation accuracy curve, or feature visualization.

Upload `figures/confusion_matrix.png` or replace this placeholder.

Fig. 1. Confusion matrix for the final classification model. Replace the image path with the exported figure from the project.

VI. DISCUSSION

Discuss what the results mean. Explain which model performed best, which classes were easiest or hardest to classify, and whether the results suggest overfitting, underfitting, or good generalization.

Also describe limitations of the project, such as dataset size, class imbalance, audio quality, limited hyperparameter search, or constraints in preprocessing.

VII. CONCLUSION

This project developed a machine learning pipeline for classifying heart, lung, and mixed audio recordings. Summarize the final model, the best performance achieved, and the main lesson learned from the project. End with possible future work, such as collecting more data, improving feature extraction, testing deeper models, or deploying the model in an application.

ACKNOWLEDGMENT

Optional: thank the instructor, course staff, dataset providers, or teammates. Delete this section if it is not needed.

REFERENCES

[1] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.
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